Mongo Query Operators

Displaying the total documents in a collection

Count the number of documents in sales collections.

```
>use sample_supplies
>show collections
```
sales

```
>db.sales.countDocuments()
```
5000

Use limit() to control number of documents to display

Display only the first 3 documents of the sales collection

```
>db.sales.find().limit(5)
```

```
[
  {
    _id: ObjectId("5bd761dcae323e45a93ccffb"),
    saleDate: ISODate("2015-11-16T22:02:34.415Z"),
    items: [
      {
        name: 'printer paper',
        tags: [ 'office', 'stationary' ],
        price: Decimal128("17.3"),
        quantity: 1
      },
      {
        name: 'binder',
        tags: [ 'school', 'general', 'organization' ],
        price: Decimal128("23.36"),
        quantity: 3
      },
      {
        name: 'envelopes',
        tags: [ 'stationary', 'office', 'general' ],
        price: Decimal128("21.53"),
        quantity: 10
      },
      {
        name: 'pens',
```

```
    tags: [ 'writing', 'office', 'school', 'stationary' ],
    price: Decimal128("57.02"),
    quantity: 4
  },
  {
    name: 'notepad',
    tags: [ 'office', 'writing', 'school' ],
    price: Decimal128("14.04"),
    quantity: 4
  },
  {
    name: 'notepad',
    tags: [ 'office', 'writing', 'school' ],
    price: Decimal128("32.2"),
    quantity: 1
  },
  {
    name: 'binder',
    tags: [ 'school', 'general', 'organization' ],
    price: Decimal128("20.2"),
    quantity: 5
  },
  {
    name: 'pens',
    tags: [ 'writing', 'office', 'school', 'stationary' ],
    price: Decimal128("37.37"),
    quantity: 4
  },
  {
    name: 'notepad',
    tags: [ 'office', 'writing', 'school' ],
    price: Decimal128("34.61"),
    quantity: 1
  },
  {
    name: 'backpack',
    tags: [ 'school', 'travel', 'kids' ],
    price: Decimal128("102.7"),
    quantity: 3
  }
],
storeLocation: 'New York',
customer: { gender: 'F', age: 39, email: 'beecho@wic.be', satisfaction: 3 },
couponUsed: false,
```

```
  purchaseMethod: 'In store'
},
{
  _id: ObjectId("5bd761dcae323e45a93cd00b"),
  saleDate: ISODate("2016-05-16T03:43:15.808Z"),
  items: [
    {
      name: 'pens',
      tags: [ 'writing', 'office', 'school', 'stationary' ],
      price: Decimal128("72.32"),
      quantity: 4
    },
    {
      name: 'laptop',
      tags: [ 'electronics', 'school', 'office' ],
      price: Decimal128("796.74"),
      quantity: 1
    },
    {
      name: 'notepad',
      tags: [ 'office', 'writing', 'school' ],
      price: Decimal128("14.38"),
      quantity: 4
    },
    {
      name: 'backpack',
      tags: [ 'school', 'travel', 'kids' ],
      price: Decimal128("158.88"),
      quantity: 5
    },
    {
      name: 'notepad',
      tags: [ 'office', 'writing', 'school' ],
      price: Decimal128("6.96"),
      quantity: 2
    }
  ],
  storeLocation: 'Seattle',
  customer: {
    gender: 'F',
    age: 26,
    email: 'kupeen@gareha.ne',
    satisfaction: 5
  },
```

```
  couponUsed: true,
  purchaseMethod: 'In store'
},
{
  _id: ObjectId("5bd761dcae323e45a93ccfee"),
  saleDate: ISODate("2014-11-11T02:13:51.893Z"),
  items: [
    {
      name: 'laptop',
      tags: [ 'electronics', 'school', 'office' ],
      price: Decimal128("604.12"),
      quantity: 4
    },
    {
      name: 'binder',
      tags: [ 'school', 'general', 'organization' ],
      price: Decimal128("11.05"),
      quantity: 7
    },
    {
      name: 'binder',
      tags: [ 'school', 'general', 'organization' ],
      price: Decimal128("20.94"),
      quantity: 5
    },
    {
      name: 'backpack',
      tags: [ 'school', 'travel', 'kids' ],
      price: Decimal128("61.16"),
      quantity: 5
    },
    {
      name: 'notepad',
      tags: [ 'office', 'writing', 'school' ],
      price: Decimal128("29.81"),
      quantity: 3
    },
    {
      name: 'printer paper',
      tags: [ 'office', 'stationary' ],
      price: Decimal128("31.19"),
      quantity: 5
    },
    {
```

```
      name: 'pens',
      tags: [ 'writing', 'office', 'school', 'stationary' ],
      price: Decimal128("47.12"),
      quantity: 1
    },
    {
      name: 'notepad',
      tags: [ 'office', 'writing', 'school' ],
      price: Decimal128("36.71"),
      quantity: 2
    },
    {
      name: 'pens',
      tags: [ 'writing', 'office', 'school', 'stationary' ],
      price: Decimal128("69.28"),
      quantity: 3
    },
    {
      name: 'notepad',
      tags: [ 'office', 'writing', 'school' ],
      price: Decimal128("14.05"),
      quantity: 1
    }
  ],
  storeLocation: 'London',
  customer: { gender: 'F', age: 40, email: 'pan@cak.zm', satisfaction: 5 },
  couponUsed: false,
  purchaseMethod: 'In store'
 }
]
```

Equality Operators

- $eq: Matches values that are equal to a specified value.
- $ne: Matches values that are not equal to a specified value.

Let us display just one document and check the data type for each field:

```
>db.sales.findOne()
{
 _id: ObjectId("5bd761dcae323e45a93ccffb"),
 saleDate: ISODate("2015-11-16T22:02:34.415Z"),
 items: [
  {
   name: 'printer paper',
```

```
  tags: [ 'office', 'stationary' ],
  price: Decimal128("17.3"),
  quantity: 1
},
{
  name: 'binder',
  tags: [ 'school', 'general', 'organization' ],
  price: Decimal128("23.36"),
  quantity: 3
},
{
  name: 'envelopes',
  tags: [ 'stationary', 'office', 'general' ],
  price: Decimal128("21.53"),
  quantity: 10
},
{
  name: 'pens',
  tags: [ 'writing', 'office', 'school', 'stationary' ],
  price: Decimal128("57.02"),
  quantity: 4
},
{
  name: 'notepad',
  tags: [ 'office', 'writing', 'school' ],
  price: Decimal128("14.04"),
  quantity: 4
},
{
  name: 'notepad',
  tags: [ 'office', 'writing', 'school' ],
  price: Decimal128("32.2"),
  quantity: 1
},
{
  name: 'binder',
  tags: [ 'school', 'general', 'organization' ],
  price: Decimal128("20.2"),
  quantity: 5
},
{
  name: 'pens',
  tags: [ 'writing', 'office', 'school', 'stationary' ],
  price: Decimal128("37.37"),
```

```
    quantity: 4
   },
   {
    name: 'notepad',
    tags: [ 'office', 'writing', 'school' ],
    price: Decimal128("34.61"),
    quantity: 1
   },
   {
    name: 'backpack',
    tags: [ 'school', 'travel', 'kids' ],
    price: Decimal128("102.7"),
    quantity: 3
   }
  ],
  storeLocation: 'New York',
  customer: { gender: 'F', age: 39, email: 'beecho@wic.be', satisfaction: 3 },
  couponUsed: false,
  purchaseMethod: 'In store'
```

Find the first 3 documents where store location is equal to New York. Display only fields store location, customer, and saleDate:

```
>db.sales.find(
{storeLocation: 'New York'},
{storeLocation: 1, customer: 1, salesDate: 1}
).
limit(3)
```

```
[
 {
  _id: ObjectId("5bd761dcae323e45a93ccffb"),
  storeLocation: 'New York',
  customer: { gender: 'F', age: 39, email: 'beecho@wic.be', satisfaction: 3 }
 },
 {
  _id: ObjectId("5bd761dcae323e45a93cd03f"),
  storeLocation: 'New York',
  customer: { gender: 'F', age: 59, email: 'kave@we.com', satisfaction: 5 }
 },
 {
  _id: ObjectId("5bd761dcae323e45a93cd09e"),
  storeLocation: 'New York',
  customer: { gender: 'M', age: 34, email: 'iz@covikse.com', satisfaction: 5 }
```

```
 }
]
```

Another version using the $eq operator:

```
> db.sales.find(
{ storeLocation:
{ $eq: 'New York' }
},
{ storeLocation: 1, customer: 1, salesDate: 1 }
)
.limit(3)
```

```
[
 {
  _id: ObjectId("5bd761dcae323e45a93ccffb"),
  storeLocation: 'New York',
  customer: { gender: 'F', age: 39, email: 'beecho@wic.be', satisfaction: 3 }
 },
 {
  _id: ObjectId("5bd761dcae323e45a93cd03f"),
  storeLocation: 'New York',
  customer: { gender: 'F', age: 59, email: 'kave@we.com', satisfaction: 5 }
 },
 {
  _id: ObjectId("5bd761dcae323e45a93cd09e"),
  storeLocation: 'New York',
  customer: { gender: 'M', age: 34, email: 'iz@covikse.com', satisfaction: 5 }
 }
]
```

It seems we have a lot of documents equal to 'New York'. Let us count them:

```
>db.sales.countDocuments(
{ storeLocation: { $eq: 'New York'}}
)
```

501

We have 5,000 documents. Therefore, we have 4,449 not equal to 'New York':

```
> db.sales.countDocuments(
{storeLocation : {$ne : 'New York'}}
)
```

4,499

Comparison Operators

- $gt: Matches values that are greater than a specified value.
- $gte: Matches values that are greater than or equal to a specified value.
- $lt: Matches values that are less than a specified value.
- $lte: Matches values that are less than or equal to a specified value.

You are required to display documents where the customer's age is equal or greater than 50. Show only storeLocation, customer, and purchased method Limit your answer to 5 documents only.

```
> db.sales.find(
{'customer.age' : {$gte : 50}},
{ customer: 1, purchaseMethod: 1, storeLocation:1}
).
limit(5)
```

```
[
 {
  _id: ObjectId("5bd761dcae323e45a93cd007"),
  storeLocation: 'Seattle',
  customer: { gender: 'M', age: 60, email: 'kege@nibomjo.gq', satisfaction: 1 },
  purchaseMethod: 'In store'
 },
 {
  _id: ObjectId("5bd761dcae323e45a93cd00a"),
  storeLocation: 'Denver',
  customer: {
   gender: 'M',
   age: 66,
   email: 'henbaw@dasekoh.bf',
   satisfaction: 4
  },
  purchaseMethod: 'Online'
 },
 {
  _id: ObjectId("5bd761dcae323e45a93cd012"),
  storeLocation: 'London',
  customer: { gender: 'F', age: 74, email: 'wapunu@ba.bm', satisfaction: 1 },
  purchaseMethod: 'Online'
 },
```

```
  {
    _id: ObjectId("5bd761dcae323e45a93cd021"),
    storeLocation: 'London',
    customer: { gender: 'F', age: 53, email: 'we@bedijohaz.gp', satisfaction: 3 },
    purchaseMethod: 'In store'
  },
  {
    _id: ObjectId("5bd761dcae323e45a93cd025"),
    storeLocation: 'Denver',
    customer: { gender: 'M', age: 57, email: 'hegbif@lun.to', satisfaction: 4 },
    purchaseMethod: 'Online'
  }
]
```

Logical Operators

- $and: Joins query clauses with a logical AND.
- $or: Joins query clauses with a logical OR.
- $not: Inverts the effect of a query expression.

You are required to display documents where the customer's age is between 40 and 50. Show only storeLocation, customer, and purchased method. Limit your answer to 5 documents only.

```
> db.sales.find( { $and :
[{ 'customer.age': {$gte :40}},
{'customer.age' : {$lte : 50}}
]
},
{storeLocation: 1, purchaseMethod: 1, customer: 1})
.limit(5)
```

```
[
  {
    _id: ObjectId("5bd761dcae323e45a93cd007"),
    storeLocation: 'Seattle',
    customer: { gender: 'M', age: 60, email: 'kege@nibomjo.gq', satisfaction: 1 },
    purchaseMethod: 'In store'
  },
  {
    _id: ObjectId("5bd761dcae323e45a93cd00a"),
    storeLocation: 'Denver',
    customer: {
      gender: 'M',
      age: 66,
      email: 'henbaw@dasekoh.bf',
```

```
    satisfaction: 4
   },
   purchaseMethod: 'Online'
  },
  {
   _id: ObjectId("5bd761dcae323e45a93cd021"),
   storeLocation: 'London',
   customer: { gender: 'F', age: 53, email: 'we@bedijohaz.gp', satisfaction: 3 },
   purchaseMethod: 'In store'
  }
]
```

Show documents whose age is less than 30 or the purchase method is 'Online'. Limit your answer in 3 documents only.

```
>db.sales.find(
{$or : [
{ purchaseMethod : 'Online'}, {'customer.age' : {$lte : 30}}
]
},
{storeLocation: 1, purchaseMethod: 1, customer: 1}
)
.limit(3)
[
 {
   _id: ObjectId("5bd761dcae323e45a93cd00b"),
   storeLocation: 'Seattle',
   customer: {
    gender: 'F',
    age: 26,
    email: 'kupeen@gareha.ne',
    satisfaction: 5
   },
   purchaseMethod: 'In store'
 },
 {
   _id: ObjectId("5bd761dcae323e45a93ccffc"),
   storeLocation: 'Denver',
   customer: { gender: 'M', age: 30, email: 'ziguh@lu.bf', satisfaction: 4 },
   purchaseMethod: 'In store'
 },
 {
   _id: ObjectId("5bd761dcae323e45a93cd00a"),
   storeLocation: 'Denver',
```

```
    customer: {
      gender: 'M',
      age: 66,
      email: 'henbaw@dasekoh.bf',
      satisfaction: 4
    },
    purchaseMethod: 'Online'
  }
]
```

Array Operators

- $in: Matches any of the values specified in an array.
- $nin: Matches none of the values specified in an array.
- $all: Matches all of the values specified in an array.

Find first 3 documents where items tags includes office, general, school

```
> db.sales.find(
{ 'items.tags': { $in: ['school', 'general', 'office'] } },
{ customer: 1, storeLocation: 1, purchaseMethod: 1, items : 1 }).
limit(3)
```

```
[
  {
    _id: ObjectId("5bd761dcae323e45a93ccffb"),
    items: [
      {
        name: 'printer paper',
        tags: [ 'office', 'stationary' ],
        price: Decimal128("17.3"),
        quantity: 1
      },
      {
        name: 'binder',
        tags: [ 'school', 'general', 'organization' ],
        price: Decimal128("23.36"),
        quantity: 3
      },
      {
        name: 'envelopes',
        tags: [ 'stationary', 'office', 'general' ],
        price: Decimal128("21.53"),
        quantity: 10
      },
```

```
  {
    name: 'pens',
    tags: [ 'writing', 'office', 'school', 'stationary' ],
    price: Decimal128("57.02"),
    quantity: 4
  },
  {
    name: 'notepad',
    tags: [ 'office', 'writing', 'school' ],
    price: Decimal128("14.04"),
    quantity: 4
  },
  {
    name: 'notepad',
    tags: [ 'office', 'writing', 'school' ],
    price: Decimal128("32.2"),
    quantity: 1
  },
  {
    name: 'binder',
    tags: [ 'school', 'general', 'organization' ],
    price: Decimal128("20.2"),
    quantity: 5
  },
  {
    name: 'pens',
    tags: [ 'writing', 'office', 'school', 'stationary' ],
    price: Decimal128("37.37"),
    quantity: 4
  },
  {
    name: 'notepad',
    tags: [ 'office', 'writing', 'school' ],
    price: Decimal128("34.61"),
    quantity: 1
  },
  {
    name: 'backpack',
    tags: [ 'school', 'travel', 'kids' ],
    price: Decimal128("102.7"),
    quantity: 3
  }
],
storeLocation: 'New York',
```

```
  customer: { gender: 'F', age: 39, email: 'beecho@wic.be', satisfaction: 3 },
  purchaseMethod: 'In store'
},
{
  _id: ObjectId("5bd761dcae323e45a93cd00b"),
  items: [
    {
      name: 'pens',
      tags: [ 'writing', 'office', 'school', 'stationary' ],
      price: Decimal128("72.32"),
      quantity: 4
    },
    {
      name: 'laptop',
      tags: [ 'electronics', 'school', 'office' ],
      price: Decimal128("796.74"),
      quantity: 1
    },
    {
      name: 'notepad',
      tags: [ 'office', 'writing', 'school' ],
      price: Decimal128("14.38"),
      quantity: 4
    },
    {
      name: 'backpack',
      tags: [ 'school', 'travel', 'kids' ],
      price: Decimal128("158.88"),
      quantity: 5
    },
    {
      name: 'notepad',
      tags: [ 'office', 'writing', 'school' ],
      price: Decimal128("6.96"),
      quantity: 2
    }
  ],
  storeLocation: 'Seattle',
  customer: {
    gender: 'F',
    age: 26,
    email: 'kupeen@gareha.ne',
    satisfaction: 5
  },
```

```
  purchaseMethod: 'In store'
},
{
  _id: ObjectId("5bd761dcae323e45a93ccfee"),
  items: [
    {
      name: 'laptop',
      tags: [ 'electronics', 'school', 'office' ],
      price: Decimal128("604.12"),
      quantity: 4
    },
    {
      name: 'binder',
      tags: [ 'school', 'general', 'organization' ],
      price: Decimal128("11.05"),
      quantity: 7
    },
    {
      name: 'binder',
      tags: [ 'school', 'general', 'organization' ],
      price: Decimal128("20.94"),
      quantity: 5
    },
    {
      name: 'backpack',
      tags: [ 'school', 'travel', 'kids' ],
      price: Decimal128("61.16"),
      quantity: 5
    },
    {
      name: 'notepad',
      tags: [ 'office', 'writing', 'school' ],
      price: Decimal128("29.81"),
      quantity: 3
    },
    {
      name: 'printer paper',
      tags: [ 'office', 'stationary' ],
      price: Decimal128("31.19"),
      quantity: 5
    },
    {
      name: 'pens',
      tags: [ 'writing', 'office', 'school', 'stationary' ],
```

```
      price: Decimal128("47.12"),
      quantity: 1
    },
    {
      name: 'notepad',
      tags: [ 'office', 'writing', 'school' ],
      price: Decimal128("36.71"),
      quantity: 2
    },
    {
      name: 'pens',
      tags: [ 'writing', 'office', 'school', 'stationary' ],
      price: Decimal128("69.28"),
      quantity: 3
    },
    {
      name: 'notepad',
      tags: [ 'office', 'writing', 'school' ],
      price: Decimal128("14.05"),
      quantity: 1
    }
  ],
  storeLocation: 'London',
  customer: { gender: 'F', age: 40, email: 'pan@cak.zm', satisfaction: 5 },
  purchaseMethod: 'In store'
 }
]
```

Sort documents

Find the first 5 documents where the store location is in 'New York'. Show only purchase method, customer, and store location. Sort according to the customer's age in **descendin**g order.

```
> db.sales.find(
... { storeLocation : 'New York' },
... { customer : 1, storeLocation : 1, purchaseMethod: 1}
... ).sort( { 'customer.age' : -1}).limit(5)
[
 {
  _id: ObjectId("5bd761dcae323e45a93ccfe8"),
  storeLocation: 'Denver',
  customer: {
    gender: 'M',
```

```
      age: 42,
      email: 'cauho@witwuta.sv',
      satisfaction: 4
    },
    purchaseMethod: 'Online'
  },
  {
    _id: ObjectId("5bd761dcae323e45a93ccfe9"),
    storeLocation: 'Seattle',
    customer: { gender: 'M', age: 50, email: 'keecade@hem.uy', satisfaction: 5 },
    purchaseMethod: 'Phone'
  },
  {
    _id: ObjectId("5bd761dcae323e45a93ccfeb"),
    storeLocation: 'Seattle',
    customer: { gender: 'F', age: 45, email: 'vatires@ta.pe', satisfaction: 3 },
    purchaseMethod: 'In store'
  },
  {
    _id: ObjectId("5bd761dcae323e45a93ccfec"),
    storeLocation: 'London',
    customer: { gender: 'M', age: 40, email: 'dotzu@ib.sh', satisfaction: 4 },
    purchaseMethod: 'In store'
  },
  {
    _id: ObjectId("5bd761dcae323e45a93ccfed"),
    storeLocation: 'London',
    customer: { gender: 'M', age: 44, email: 'owtar@pu.cd', satisfaction: 2 },
    purchaseMethod: 'In store'
  }
]
Atlas atlas-13i2gn-shard-0 [primary] sample_supplies> db.sales.find(
... { storeLocation : 'New York' },
... { customer : 1, storeLocation : 1, purchaseMethod: 1}
... ).sort( { 'customer.age' : -1}).limit(5)
[
  {
    _id: ObjectId("5bd761deae323e45a93cdf3d"),
    storeLocation: 'New York',
    customer: { gender: 'F', age: 75, email: 'ne@potadto.vg', satisfaction: 4 },
    purchaseMethod: 'In store'
  },
  {
    _id: ObjectId("5bd761dcae323e45a93cd144"),
```

```
    storeLocation: 'New York',
    customer: {
      gender: 'M',
      age: 75,
      email: 'jiwdivav@ujcanto.tw',
      satisfaction: 4
    },
    purchaseMethod: 'Online'
  },
  {
    _id: ObjectId("5bd761ddae323e45a93cd4b8"),
    storeLocation: 'New York',
    customer: { gender: 'F', age: 75, email: 'fok@ro.us', satisfaction: 5 },
    purchaseMethod: 'In store'
  },
  {
    _id: ObjectId("5bd761deae323e45a93cde6a"),
    storeLocation: 'New York',
    customer: {
      gender: 'M',
      age: 74,
      email: 'nufke@naamete.at',
      satisfaction: 4
    },
    purchaseMethod: 'In store'
  },
  {
    _id: ObjectId("5bd761ddae323e45a93cd72e"),
    storeLocation: 'New York',
    customer: { gender: 'M', age: 74, email: 'gu@sajkahe.pe', satisfaction: 2 },
    purchaseMethod: 'In store'
  }
]
```